

Mohammad Khalid Hasan Nabil

Dhaka, Bangladesh

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OBJECTIVE

Undergraduate Information and Communication Engineering student (5th semester) with a strong foundation in scientific and physics-informed machine learning. Seeking a Research Assistantship or a fully-funded Master's opportunity in the UAE or Saudi Arabia. Proven experience in surrogate modeling, environmental forecasting, and deep learning across cross-disciplinary engineering applications.

EDUCATION

BSc in Information and Communication Engineering

Bangladesh University of Professionals (BUP)

CGPA: 3.38/4.00

Expected 2028

Current Status: 5th Semester

Higher Secondary Certificate (HSC)

Dhaka Residential Model College

Dec 2021

GPA: 5.00/5.00

Secondary School Certificate (SSC)

Savar Cantonment Morning Glory School & College

July 2019

GPA: 5.00/5.00

RESEARCH & WORK EXPERIENCE

Medical Image Processing Research (Undergraduate Thesis)

Ongoing

- Applying deep learning tools and Knowledge Distillation to medical image analysis under the supervision of Md Appel Mahmud Pranto.

AI Demodulator Development (Communication Laboratory, BUP)

Ongoing

- Building a Physics Guided Neural Network (PGNN) demodulator for digital modulation schemes, including BPSK, QPSK and 8-QAM.
- Collecting and processing signal data and hardware interference simulations via the Switch Click method.

Environmental Forecasting & Climate Modeling

2025 – Present

- Engineered a Python script utilizing weather APIs and pandas to compile historical climate features across all 64 districts in Bangladesh.
- Built a Fourier-guided ML framework combining signal-based features with ensemble learning to predict 8 Temperature Correlated Conditions (TCC).
- Calculated the Water Quality Index of rivers in Bangladesh using hybrid Random Forest and LSTM time-series validation models.

Physics-Guided Machine Learning in Jet Engines

Published April 2026

- Modeled physical systems for bird-strike analysis on rotating jet-engine blades.
- Conducted a comparative study of Lagrangian and Smoothed Particle Hydrodynamics (SPH) simulations to solve complex physical equations.

Tensile Strength Prediction in Additive Manufacturing

- Applied physics-augmented machine learning to predict the ultimate tensile strength of 3D-printed parts from print parameters.
- Contributed to data analysis and manuscript drafting on infill design influences.

PUBLICATIONS & CONFERENCES

- **Journal:** M. K. H. Nabil, J. A. Sajid, I. Grgić, J. Marijić, S. B. Rayhan. "Physics-Guided Machine Learning Surrogates for Bird Strike Analysis on Rotating Jet Engine Blades Through a Comparative Study of Lagrangian and SPH Simulations." *Modelling (MDPI)*, Vol. 7, Issue 3, p. 80, April 2026.

- **Journal (Under Review):** M. K. H. Nabil, T. Parsha, M. H. Usama, A. Akter, I. Grgić, M. A. Moni. "Projections of Heat-Stress Conditions Across Bangladesh Using Explainable Machine Learning and Climate Trend Integration." *Climatic Change (Springer Nature)*.
- **Conference:** M. K. H. Nabil, I. Grgić. "A Fourier-Guided Machine Learning Framework for District-Level Dew Point Comfort Forecasting Across 64 districts of Bangladesh (2025–2027)." *August 2026*.
- **Conference:** L. Brčić, I. Grgić, J. Marijić, M. Karakašić, M. K. H. Nabil. "Analysis of the Influence of Infill Density and Infill Geometry on the Tensile Strength of PLA Structures Filled with Photopolymer Resin." *August 2026*.

EXTRACURRICULAR & LEADERSHIP

Team Lead, 42nd VFS Annual Student Design Competition(Participitated) 2024 – 2025

- Led a 7-member interdisciplinary team to design a hydrogen-powered VTOL aircraft.
- Selected and integrated a PEMFC powertrain based on the Toyota Mirai's architecture into a hybridized Guimbal Cabri G2/Robinson R22 concept.

Sports Achievements

- Champion, Bangabandhu Memorial Cricket League, Savar (Aug 2023)
- Champion, Inter-Department Cricket Tournament (Dec 2018)
- Champion, Inter-Hall Badminton Tournament (July 2018)

RECOMMENDATIONS

Dr. Ivan Grgić

- Mechanical Engineering Faculty in Slavonski Brod, University of Slavonski Brod, Slavonski Brod, 35000, Croatia.
- e-mail: igrgic@unisb.hr

Dr. Ali Moni

- Artificial Intelligence & Data Science, School of Health and Rehabilitation Sciences, Faculty of Health and Behavioural Sciences, The University of Queensland St Lucia, Australia
- e-mail: m.moni@uq.edu.au

Dr. MD Nasir Uddin

- Bangladesh University of Professionals, Faculty of Science and Technology, Dhaka, Bangladesh.
- e-mail: igrgic@unisb.hr

Dr. MD Siddiqur Rahman

- Department of Statistics, Jagannath University, Dhaka, Bangladesh.
- e-mail: rsiddiq11@stat.jnu.ac.bd